Kawasaki Supercharged H2 Engine Family

A TECHNICAL REFERENCE FOR CAR INSTALLATIONS

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General guidance only. Not a specification or build manual. Verify all figures against the individual engine before relying on them.

PLEASE READ FIRST

This document is **general guidance only**. It is published to help owners, builders and prospective buyers understand the Kawasaki supercharged H2 engine family and what is involved in running one outside its original motorcycle.

It is not a specification, a build manual, or advice on any individual engine or installation. Specifications vary by model year, market and production revision, and engines that have previously been modified or rebuilt may differ from anything described here. Any figure quoted should be verified against the specific engine in front of you before it is relied upon.

Installing a motorcycle engine in a car is a substantial engineering exercise with real consequences if it is done badly. Nothing here replaces proper inspection, measurement and calibration by a competent person.

If you would like advice on a particular engine, car or installation, please get in touch — that conversation is free, and considerably more useful than a general document.

Purpose and Scope

The H2 family is frequently discussed as though it were one engine. It is not.

The variants share a common architecture and a common design philosophy, but the differences between them are more substantial than is generally assumed — and several of them are mechanical rather than electronic. Those differences determine how an engine should be boosted, fuelled and calibrated once the original motorcycle electronics are removed, which in a car installation they always are.

This document is written for **car installations generally**. The H2 engine has found its way into lightweight sports cars, kit and component cars, sports racing cars, single-seaters, hillclimb specials, buggies and one-off specialist chassis. The packaging, cooling and transmission solutions differ enormously between them. What does not differ is the underlying behaviour of the engine, its lubrication requirements, and its need for a properly specified standalone electrical and management system — and those are the subjects covered here.

Manufacturer figures are Kawasaki-published or press-released values. Where something derives from our own conversion and development work, it is identified as such.

Platform Overview

Kawasaki introduced the supercharged H2 platform in 2015. Every engine in the family is a 998 cc inline-four with a centrifugal supercharger developed in-house by Kawasaki Heavy Industries, drawing on the group's gas turbine and aerospace engineering.

These engines were designed around forced induction from the outset. They are not naturally aspirated engines with a blower added, and the architecture, piston cooling and lubrication strategy all reflect that. It is one of the reasons the platform tolerates car use as well as it does — provided its actual limitations are respected.

COMMON TO ALL VARIANTS	SPECIFICATION
Configuration	998 cc inline four-cylinder, DOHC, 16-valve
Bore × stroke	76.0 mm × 55.0 mm
Induction	Centrifugal supercharger, in-house Kawasaki design
Supercharger drive	Planetary gear train from the crankshaft via an intermediary gear, stepping impeller speed up to 9.2× crank speed
Impeller	69 mm, forged aluminium, 5-axis CNC machined
Maximum impeller speed	Approaching 130,000 rpm
Charge cooling	None fitted as standard — no intercooler
Piston cooling	Oil jet beneath each piston
Transmission	6-speed, chain final drive
Lubrication (standard)	Wet sump

> Kawasaki's position is that the supercharger design does not raise charge temperature sufficiently to require an intercooler on the motorcycle. That reasoning is sound in its original application. **It does not automatically transfer to a car**, where airflow through the engine bay, ambient recirculation, sustained load and heat soak in traffic or on a grid can all be considerably worse than anything a motorcycle sees.

Variant Guide

MODEL	ROLE	COMPRESSION	CLAIMED POWER	PEAK TORQUE
Ninja H2 2015–2018	Road-legal hypersport	8.5:1	200 PS (210 PS ram air) @ 11,000 rpm	133.5 Nm
Ninja H2 2019 on	Road-legal hypersport	8.5:1	231 PS (242 PS ram air) @ 11,500 rpm	141.7 Nm
Ninja H2R	Track only	8.3:1	310 PS (326 PS ram air) @ 14,000 rpm	156 Nm @ 12,500 rpm
Ninja H2 SX / SX SE	Sport touring	11.2:1	200 PS (210 PS ram air)	137 Nm @ 9,500 rpm
Z H2 / Z H2 SE	Naked roadster	11.2:1	200 PS @ 11,000 rpm	137 Nm @ 8,500 rpm

The 2019 Ninja H2 uplift came from adopting the H2 SX air filter, intake chamber, spark plugs and ECU. Notably, Kawasaki did **not** adopt the H2 SX supercharger. The H2 SX engine itself did not receive the 2019 increase and has remained substantially unchanged since its 2018 introduction.

What Actually Differs Between the Variants

This is the section most worth reading carefully, because the common assumption — that these engines differ only in airbox and mapping — is wrong.

Two distinct engine specifications, not one

The family splits into two meaningfully different builds.

Low compression, high boost — Ninja H2 and H2R (8.5:1 / 8.3:1)

Built for peak output. More aggressive camshafts, particularly on the H2R. Higher boost. Greater tolerance of raised boost, because the knock threshold sits further away.

High compression, moderate boost — Ninja H2 SX and Z H2 (11.2:1)

Built for real-world torque and efficiency. These use a different cylinder head, different camshafts, different crankshaft and different pistons from the Ninja H2, along with a different supercharger impeller and plenum chamber. The complete engine is around 3 kg lighter than the Ninja H2 unit.

The superchargers are genuinely distinct units between the two groups — not the same part with a different map. Even between the H2 SX and the Z H2 the impeller blade design differs, biasing the Z H2 further towards mid-range torque. On the high-compression engines the impeller carries six full-length blades at the tip with twelve at the base.

Why this matters in a car

Compression ratio is the dominant variable once you are running your own calibration:

- **Boost ceiling.** An 11.2:1 engine reaches its knock limit at markedly less boost than an 8.5:1 engine. If the objective is a high-boost, high-output build, the low-compression Ninja H2 specification is the more tolerant starting point.
- **Fuel sensitivity.** The high-compression engines are more sensitive to fuel quality and to charge temperature — which, as noted above, is a harder problem in a car than on a bike.
- **Off-boost behaviour.** The high-compression engines are considerably better below and between boost. In a car — heavier than a motorcycle, and usually geared longer — this matters far more than it does on the original bike. Torque arriving 1,000 rpm earlier on the Z H2 than the H2 SX is a real driveability advantage in a road or hillclimb car.
- **Calibration is not transferable.** Ignition and fuelling developed on one compression ratio should never be carried across to the other.

There is no single "best" variant. There is a variant that suits the intended use, and choosing it deliberately is worth considerably more than choosing on price or availability.

Transmission note

The Z H2 uses a dog-ring transmission. Where the original gearbox is retained in a car installation — as it commonly is — this affects shift feel, shift actuation design and expectations about shift quality at low speed. It is worth knowing before the transmission solution is designed around it.

Identifying an Engine Correctly

Do not identify an H2 engine by its engine number prefix.

The **ZXT00NE** prefix appears across the supercharged family — on Z H2, Ninja H2 SX and Ninja H2 units alike. It identifies the 998 cc supercharged engine family, not the compression ratio or state of tune. Serial numbers run sequentially across the family, so a number falling within a known range proves nothing either.

Given that the two engine groups differ in head, cams, crank, pistons and supercharger, buying on prefix alone is a genuine risk.

PREFIX	MODEL FAMILY
...ZRT00...	Z H2 (ZR1000 series)
...ZXT02...	Ninja H2 SX family
ZX1002... (model code)	Ninja H2, 2019 on

Where an engine has been separated from its motorcycle and no VIN is available, establish the specification physically or by part number before any calibration work begins. Cosmetic finish is not a reliable identifier — supercharger housing colour varies by model and year and is not a Kawasaki specification code.

Lubrication in Car Installations

Why the standard wet sump is the limiting factor

The H2 wet sump is well designed for its intended purpose. A motorcycle leans into a corner, so the oil surface stays broadly perpendicular to the sump and the pickup remains submerged. Sustained lateral loading is simply not a design case.

A car does not lean. The same corner now produces sustained lateral acceleration acting across the sump, and oil migrates away from the pickup and stays there for the duration of the corner. Braking and acceleration add longitudinal displacement on top. The result is intermittent pickup uncovering under precisely the conditions that also produce peak bearing load.

The severity scales with the car. A road-biased build on modest tyres will see far less of it than a light car on slicks sustaining high lateral g through long corners — which is exactly the sort of car this engine tends to end up in. Long-radius corners and banked or high-speed circuits are the worst case, because the loading is sustained rather than transient.

This is a geometry problem, not a capacity problem. It is not reliably solved by baffling, by a deeper pan, or by simply running more oil within the original sump architecture. Oil starvation failures in car-installed H2 engines are a recognised and recurring pattern wherever the original wet sump arrangement has been retained or only lightly modified.

Dry sump conversion — our approach

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FEATURE	SPECIFICATION
System type	Custom dry sump
Total system oil volume	Approximately 6 litres
Feed lines	-12 AN
Return / scavenge lines	-12 AN
Scavenge pump	Derived from a proven Suzuki Hayabusa arrangement
Oil pump gears	Uprated gears (Extreme Creations)
Pressure regulation	External regulation, OEM relief valve retained
Piston oil jets	Cleaned and flow-checked, not modified

The combination of a dry sump, substantially increased system capacity and uprated pump gearing delivers materially better lubrication security than the standard motorcycle configuration, under conditions the original system was never designed to meet. It also removes the sump depth from the packaging problem, which in a low car is frequently useful in its own right.

Oil Pressure Characteristics

The H2 runs lower oil pressure than several comparable engines — the Suzuki Hayabusa being the usual point of comparison. **This is by design and is not a fault.** The H2 lubrication strategy prioritises flow and piston cooling over headline pressure, and on the high-compression variants the pump is deliberately turned more slowly to reduce parasitic loss.

CONDITION (HOT)	TYPICAL OIL PRESSURE
Idle	~2.0 bar
Normal running	~3.0–3.5 bar
Peak rpm	~4.0–4.5 bar

> **A note on instrumentation.** Motorsport pressure sensors frequently indicate apparently low or delayed readings on these engines, because the available sensor location sits downstream in the oil galleries rather than at the pump outlet. Readings should be interpreted against that, not against expectations formed on a different engine family. More than one perfectly healthy H2 has been diagnosed with a problem it did not have.

Engine Speed and Calibration

Factory rev limits vary by variant. The Ninja H2 and H2R run to approximately 14,000 rpm. The Z H2 and H2 SX are limited lower, in the region of 12,000 rpm, consistent with their torque-biased role and higher compression ratio.

In a car installation the OEM loom and ECU are removed entirely, so factory limits are not inherited. Engine speed becomes a deliberate calibration decision rather than an inherited constraint.

TOO FAST TO RACE DEVELOPMENT WORK

Our own Z H2-based installation is calibrated to 14,000 rpm. The reasoning is that the rotating assembly architecture is common across the family and demonstrably operates at that speed in factory Ninja H2 form — mean piston speed at 14,000 rpm on the 55 mm stroke is approximately 25.7 m/s, a figure Kawasaki already sanctions on this bottom end. The meaningful constraint on the high-compression variants is therefore combustion and thermal loading rather than the mechanical capability of the crank and rods, and that is managed through boost, fuelling and ignition strategy.

> This is a documented position on one development engine, arrived at with supporting lubrication and cooling changes already in place. **It should not be read as a general recommendation.** Where an engine is close to standard, on OEM rods, and without the supporting systems to match, the sensible default is to build margin rather than chase the number.

Loom and Engine Management

Car installation requires complete removal of the motorcycle wiring loom and ECU. This is not optional, and the reasons are worth understanding before a budget is set:

- **Immobiliser and security integration.** The factory system expects to see motorcycle-specific components before it will permit the engine to run.

- **Ride-by-wire throttle.** Throttle actuation must be correctly reproduced and calibrated by the replacement system.
- **Crank and cam trigger interpretation.** The replacement ECU must be configured for the engine's trigger arrangement.
- **Sensor set and connector standards.** Motorcycle sensors and connectors are not always appropriate to an automotive environment, or to a car's vibration, heat and moisture exposure.
- **Supporting systems.** Cooling fan control, fuel pump control, dash, warning strategy and data logging all require deliberate provision that the bike loom will not give you.

This is where a large proportion of H2 conversion projects stall. A purpose-designed loom and a correctly configured standalone ECU remove most of that difficulty — and they remove it at the start of the project, rather than after the engine is already in the car.

Considerations When Buying an Engine

1. **Establish which variant it is** — by VIN or model code, not engine prefix or appearance. Confirm the compression ratio.
2. **Match the variant to the intended use.** High boost and high output favour the low-compression Ninja H2 specification. Strong mid-range and driveability on pump fuel favour the 11.2:1 Z H2 or H2 SX.
3. **Mileage and service history**, particularly oil change discipline.
4. **Evidence of previous car installation.** An engine previously run in a car on a wet sump, or a lightly modified sump, warrants close inspection of bearings and piston oil jets before anything else is spent on it.
5. **Supercharger condition.** The unit is not user-serviceable and is expensive to replace.
6. **Completeness.** Missing ancillaries are frequently harder to source than the engine itself.
7. **Plan the lubrication and electrical systems before purchase, not after.** Between them they account for most of the cost, most of the difficulty, and nearly all of the failures.

Document Notes

Manufacturer specifications have been checked against Kawasaki-published and press-released figures. Where a statement derives from our own development work rather than manufacturer data, this is identified in the text.

This document will be revised as the platform and our own development work progress. For advice on engine selection, dry sump conversion, loom and ECU specification, or complete installation support, contact Too Fast To Race.